

Data Engineering Track

HACKATHON 1

PROGRAM EXECUTION PLAN

Friday, 05th June, 2026

The UNOX Hackathon is a structured, hands-on problem-solving event where participants apply their analytical, SQL, and programming skills to tackle real-world challenges. The event emphasizes **team collaboration, problem-solving, and technical skill development** in the domains of **SQL Queries, Business Insights and Data Visualization**.

Overview

Team Structure

Each team will consist of **four members**, ensuring a balanced distribution of skills and fostering collaboration.

One team member will be designated as the **Scrum Master**, to take responsibility for the team's performance. Teams will mutually assign each **Backlog Item (problem statement)** through discussion, ensuring fair distribution of tasks.

Hackathon Parts

The Hackathon will have two parts:

Hackathon 1 (Mandatory)

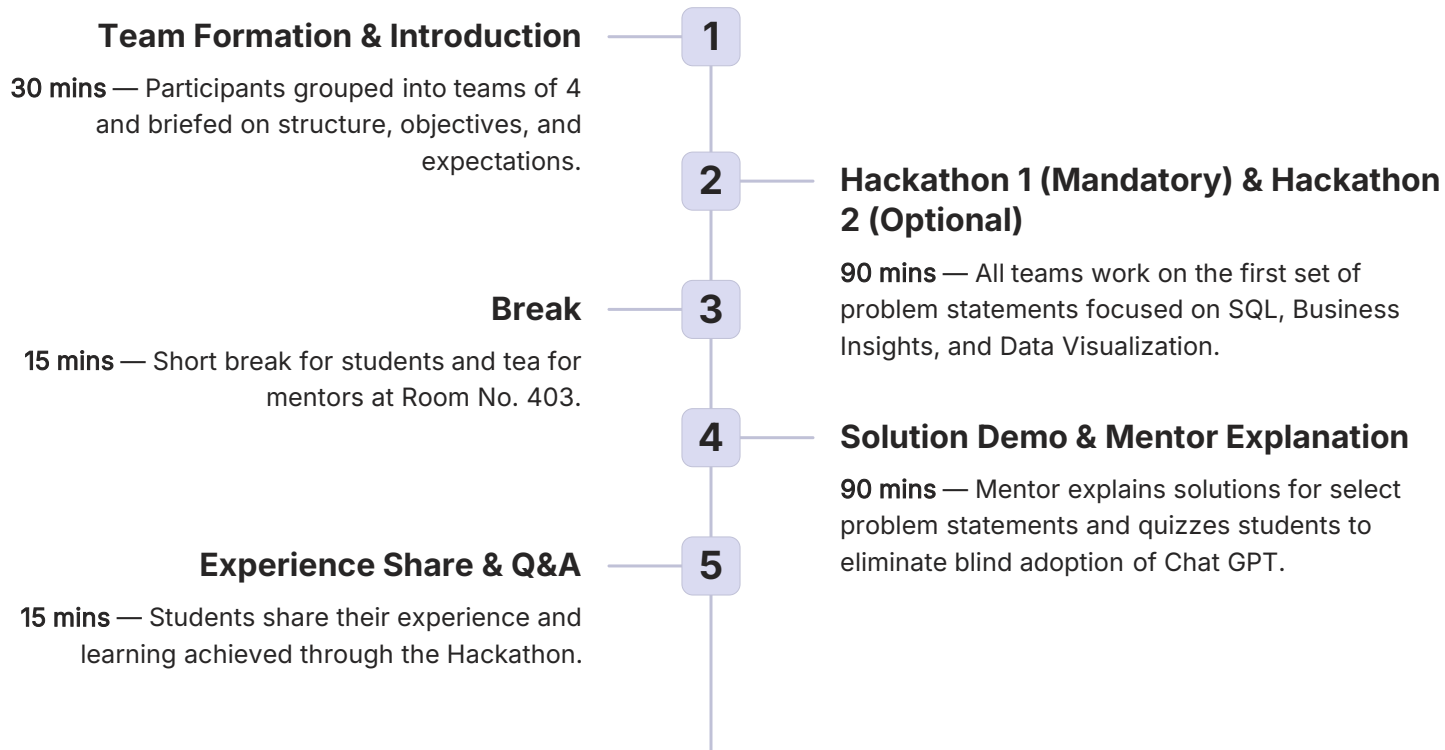
The primary set of problem statements that all teams must complete.

Hackathon 2 (Optional)

An additional set of problem statements for students who have completed the mandatory sets and wish to take on extra challenges.

Hackathon Structure & Timeline

The event will run for **240 minutes (4 hours)** and follow a structured approach:



Problems Distribution

The hackathon problem statements are structured into four tables, covering different skill areas.

Table 1: First Set of Problems (Mandatory for All Teams)

This set consists of **SQL Queries** and **Business Insights** problems. These problems are **common for all teams**.

Category	Simple	Medium	Complex	Total Problems	Max Duration (Per Problem in Minutes)	Items Per Team Member	Total Time Allocation
SQL Queries		11	5	16	15	4	60
Business Insights		6	2	12	15	2	30
Total Problems		17	7	24	—	6	90 mins



SQL Queries: Focus on retrieving and processing data efficiently.



Business Insights: Interpret data trends to derive actionable insights.

Total Time Allocation: **90 mins**

Hackathon 1 Backlog — Data Queries

The following are the mandatory SQL / Data Query problem statements at the **Medium** difficulty level:

Sl No	Problem Statement	Type	Level
1	List All Seats in a Specific Screen	Data Queries	M1
2	Count the Number of Reviews for Each Movie	Data Queries	M2
3	Fetch All Upcoming Shows	Data Queries	M3
4	Find the Most Expensive Food Items	Data Queries	M4
5	Find the Most Frequent Movie Genres	Data Queries	M5
6	Get the Most Popular Shows	Data Queries	M6
7	Find the Customers Who Have Made the Most Spending	Data Queries	M7
8	Find the Most Popular Showtimes	Data Queries	M8
9	Calculate the Average Spending Per Customer	Data Queries	M9
10	Find the Most Popular Food Items Ordered	Data Queries	M10
11	Identify Customers with the Highest Spending on Food Items	Data Queries	M11
12	Determine the Most Frequent Moviegoers	Data Queries	C1
13	Find the Most Active Booking Hours	Data Queries	C2
14	Identify the Customers Who Book the Most Tickets	Data Queries	C3
15	Get the Most Popular Movie Genres by Ticket Sales	Data Queries	C4
16	Identify the Peak Hours for Food Orders	Data Queries	C5
17	Find the Least Popular Movies	Business Insights	M1
18	Get the Most Profitable Days for Ticket Sales	Business Insights	M2
19	Identify the Customers Who Frequently Cancel Bookings	Business Insights	M3
20	Find the Most Popular Seats in Theaters	Business Insights	M4
21	Get the Average Revenue Per Booking	Business Insights	M5
22	Determine the Top Spending Customers by Booking and Food	Business Insights	M6
23	Find the Customers Who Book Tickets and Never Show Up	Business Insights	C1
24	Forecast Future User Growth Based on Historical Data	Business Insights	C2

Table 2: Second Set of Problems (Optional)

Category	Simple	Medium	Complex	Total Problems
SQL Queries			3	3
Business Insights				2
Data Visualization		4		4
Total Problems		4	3	7

Hackathon 2 Backlog — Optional Challenges

These problem statements can be taken up by students who have completed Hackathon 1 (if they have the time). The rest of the students can take them up as homework.

Sl No	Problem Statement	Hackathon	Type	Level
25	Find the Peak Revenue Time Slots for Ticket Sales	H2	Data Queries	C1
26	Identify Customers Who Purchased Both Tickets and Food	H2	Data Queries	C2
27	Find the Most Profitable Screens Based on Ticket Sales	H2	Data Queries	C3
28	Screen Performance by Booking Volume	H2	Data Visualization	M1
29	Booking Delivery Method Distribution	H2	Data Visualization	M2
30	Movie Revenue Analysis by Genre	H2	Data Visualization	M3
31	Monthly Ticket Sales Trend	H2	Data Visualization	M4

Key Takeaways

This **detailed execution plan** ensures a **structured, engaging, and productive** hackathon experience for all participants.



SQL & Data Analysis

Hands-on experience in **SQL, Data Analysis, Business Insights, and Data Visualization** through real problem-solving scenarios drawn from a cinema booking domain.



Real-World Problem Solving

Exposure to **real-world problem-solving** in **structured query languages and programming**, simulating the kind of analytical work done in industry data teams.



Collaboration & Teamwork

Collaboration and teamwork in a competitive yet learning-driven environment, including rotating Scrum Master roles to build leadership skills across all team members.

"Students should share their experience and learning achieved through the Hackathon — problem solving, team work, collaboration, communication, peer learning, learning from the mentor, and more."

- ✔ All participants are encouraged to attempt Hackathon 2 problems as homework, even if not completed during the event. Continuous practice is key to mastering data engineering skills.